

Daily Problem Solutions Booklet

Comprehensive Step-by-Step Analytical Solutions

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Thursday 11 June 2026

1.1 Daily Limit

Limit of Trigonometric Functions

Evaluate the limit:

$$\lim_{x \rightarrow 0} \frac{\sin^3(x) \cos^3(x) - x^3 \cos(2x)}{\sin^3(x) - x^3 \cos(x)}$$

Solution: To evaluate this limit, we utilize the Maclaurin series expansions for the trigonometric functions near $x = 0$.

First, we determine the series expansions for the terms in the denominator $D(x) = \sin^3(x) - x^3 \cos(x)$. Recall the standard expansion for sine:

$$\sin(x) = x - \frac{x^3}{6} + \frac{x^5}{120} - \dots$$

Cubing this series yields:

$$\sin^3(x) = \left(x - \frac{x^3}{6} + \frac{x^5}{120} - \dots\right)^3 = x^3 - \frac{x^5}{2} + \frac{13x^7}{120} - \dots$$

For the second term in the denominator:

$$x^3 \cos(x) = x^3 \left(1 - \frac{x^2}{2} + \frac{x^4}{24} - \dots\right) = x^3 - \frac{x^5}{2} + \frac{x^7}{24} - \dots$$

Subtracting these gives the expansion for the denominator:

$$D(x) = \left(x^3 - \frac{x^5}{2} + \frac{13x^7}{120}\right) - \left(x^3 - \frac{x^5}{2} + \frac{x^7}{24}\right) = \left(\frac{13}{120} - \frac{5}{120}\right)x^7 = \frac{x^7}{15} + O(x^9)$$

Next, we find the expansion for the numerator $N(x) = \sin^3(x) \cos^3(x) - x^3 \cos(2x)$. Note that $\sin^3(x) \cos^3(x) = \left(\frac{\sin(2x)}{2}\right)^3 = \frac{1}{8} \sin^3(2x)$. Using the expansion for $\sin(2x) = 2x - \frac{(2x)^3}{6} + \frac{(2x)^5}{120} - \dots = 2x - \frac{4x^3}{3} + \frac{4x^5}{15} - \dots$, we cube it:

$$\begin{aligned} \sin^3(2x) &= (2x)^3 + 3(2x)^2 \left(-\frac{4x^3}{3}\right) + 3(2x) \left(-\frac{4x^3}{3}\right)^2 + 3(2x)^2 \left(\frac{4x^5}{15}\right) + \dots \\ &= 8x^3 - 16x^5 + \left(3(2x) \left(\frac{16x^6}{9}\right) + 3(4x^2) \left(\frac{4x^5}{15}\right)\right) + \dots \\ &= 8x^3 - 16x^5 + \left(\frac{32}{3}x^7 + \frac{16}{5}x^7\right) + \dots = 8x^3 - 16x^5 + \frac{208}{15}x^7 + \dots \end{aligned}$$

Multiplying by $\frac{1}{8}$ gives:

$$\frac{1}{8} \sin^3(2x) = x^3 - 2x^5 + \frac{26}{15}x^7 + \dots$$

For the second term in the numerator:

$$x^3 \cos(2x) = x^3 \left(1 - \frac{(2x)^2}{2} + \frac{(2x)^4}{24} - \dots\right) = x^3 - 2x^5 + \frac{2}{3}x^7 + \dots$$

Subtracting these gives the expansion for the numerator:

$$N(x) = \left(x^3 - 2x^5 + \frac{26}{15}x^7\right) - \left(x^3 - 2x^5 + \frac{2}{3}x^7\right) = \left(\frac{26}{15} - \frac{10}{15}\right)x^7 = \frac{16}{15}x^7 + O(x^9)$$

Finally, we evaluate the limit by taking the ratio of the leading terms of $N(x)$ and $D(x)$:

$$\lim_{x \rightarrow 0} \frac{N(x)}{D(x)} = \lim_{x \rightarrow 0} \frac{\frac{16}{15}x^7}{\frac{1}{15}x^7} = 16$$

Friday 12 June 2026

2.1 Daily Limit

Limit of a Ratio of Sums

Evaluate the limit:

$$\lim_{n \rightarrow \infty} \frac{\sum_{j=1}^n j^2}{\sum_{k=1}^n kn} \quad \text{where } n \in \mathbb{N}$$

Solution: To evaluate this limit, we first find closed-form expressions for the series in the numerator and the denominator.

The numerator is the sum of the squares of the first n natural numbers:

$$\sum_{j=1}^n j^2 = \frac{n(n+1)(2n+1)}{6}$$

Expanding the polynomial in the numerator, the leading term is $\frac{2n^3}{6} = \frac{n^3}{3}$.

The denominator features a sum where n acts as a constant with respect to the index k . We factor out n :

$$\sum_{k=1}^n kn = n \sum_{k=1}^n k$$

The remaining series is the sum of an arithmetic progression of the first n integers:

$$\sum_{k=1}^n k = \frac{n(n+1)}{2}$$

Substituting this back into the denominator yields:

$$n \left(\frac{n(n+1)}{2} \right) = \frac{n^2(n+1)}{2}$$

Now, substitute these closed forms back into the original limit expression:

$$\lim_{n \rightarrow \infty} \frac{\frac{n(n+1)(2n+1)}{6}}{\frac{n^2(n+1)}{2}}$$

We can simplify this compound fraction by multiplying the numerator and denominator by $\frac{6}{n(n+1)}$:

$$\lim_{n \rightarrow \infty} \frac{2n+1}{3n}$$

Dividing the numerator and the denominator by the highest power of n , which is n^1 :

$$\lim_{n \rightarrow \infty} \frac{2 + \frac{1}{n}}{3} = \frac{2+0}{3} = \frac{2}{3}$$

Thus, the limit evaluates to:

$$\frac{2}{3}$$

Saturday 13 June 2026

3.1 Daily Limit

Limit Involving the Number e

Given the limit equation:

$$\lim_{x \rightarrow 0} \left(\frac{e^x + e^{2x} + e^{3x} + \dots + e^{nx}}{n} \right)^{\frac{1}{x}} = e^{10}$$

Find the value of n .

Solution: Let the limit on the left-hand side be L . As $x \rightarrow 0$, the sum inside the parenthesis approaches n , so the base approaches $\frac{n}{n} = 1$. The exponent $\frac{1}{x}$ approaches ∞ , yielding the indeterminate form 1^∞ . To evaluate this, we take the natural logarithm of the limit:

$$\ln(L) = \lim_{x \rightarrow 0} \frac{1}{x} \ln \left(\frac{1}{n} \sum_{k=1}^n e^{kx} \right)$$

This transforms the expression into the indeterminate form $\frac{0}{0}$. We apply L'Hôpital's Rule by differentiating the numerator and the denominator with respect to x :

$$\ln(L) = \lim_{x \rightarrow 0} \frac{\frac{d}{dx} \ln \left(\frac{1}{n} \sum_{k=1}^n e^{kx} \right)}{1}$$

Using the chain rule for the derivative of the logarithm:

$$\frac{d}{dx} \ln \left(\frac{1}{n} \sum_{k=1}^n e^{kx} \right) = \frac{1}{\frac{1}{n} \sum_{k=1}^n e^{kx}} \cdot \left(\frac{1}{n} \sum_{k=1}^n k e^{kx} \right)$$

Now we evaluate this derivative at $x = 0$. For the denominator factor, the sum $\sum_{k=1}^n e^0 = n$, making it $\frac{1}{n} \cdot n = 1$. For the numerator factor, the sum evaluates to $\sum_{k=1}^n k e^0 = \sum_{k=1}^n k$. Recall the sum of the first n positive integers:

$$\sum_{k=1}^n k = \frac{n(n+1)}{2}$$

Substituting these values into our limit for $\ln(L)$:

$$\ln(L) = \frac{1}{1} \cdot \left(\frac{1}{n} \cdot \frac{n(n+1)}{2} \right) = \frac{n+1}{2}$$

Exponentiating both sides to solve for L :

$$L = e^{\frac{n+1}{2}}$$

We are given that $L = e^{10}$. By equating the exponents, we can solve for n :

$$\frac{n+1}{2} = 10 \implies n+1 = 20 \implies n = 19$$

The value of n is:

Sunday 14 June 2026

4.1 Daily Limit

Limit of a Nested Radical Function

Evaluate the limit:

$$\lim_{x \rightarrow 0} \frac{\sqrt{1 + \sqrt{x+1}} - \sqrt{2}}{x}$$

Solution: Direct substitution of $x = 0$ yields $\frac{0}{0}$, an indeterminate form. To evaluate this limit analytically, we multiply the numerator and the denominator by the conjugate of the numerator, $\sqrt{1 + \sqrt{x+1}} + \sqrt{2}$:

$$\lim_{x \rightarrow 0} \frac{(\sqrt{1 + \sqrt{x+1}} - \sqrt{2})(\sqrt{1 + \sqrt{x+1}} + \sqrt{2})}{x(\sqrt{1 + \sqrt{x+1}} + \sqrt{2})}$$

Using the difference of squares identity $(a - b)(a + b) = a^2 - b^2$, the numerator simplifies to:

$$(1 + \sqrt{x+1}) - 2 = \sqrt{x+1} - 1$$

The limit expression now becomes:

$$\lim_{x \rightarrow 0} \frac{\sqrt{x+1} - 1}{x(\sqrt{1 + \sqrt{x+1}} + \sqrt{2})}$$

We still face a $\frac{0}{0}$ indeterminate form, so we rationalize the new numerator by multiplying the top and bottom by its conjugate, $\sqrt{x+1} + 1$:

$$\lim_{x \rightarrow 0} \frac{(\sqrt{x+1} - 1)(\sqrt{x+1} + 1)}{x(\sqrt{1 + \sqrt{x+1}} + \sqrt{2})(\sqrt{x+1} + 1)}$$

Applying the difference of squares again, the new numerator simplifies to:

$$(x + 1) - 1 = x$$

Substituting this back into the limit yields:

$$\lim_{x \rightarrow 0} \frac{x}{x(\sqrt{1 + \sqrt{x+1}} + \sqrt{2})(\sqrt{x+1} + 1)}$$

Since $x \rightarrow 0$ but $x \neq 0$, we can cancel the common factor of x :

$$\lim_{x \rightarrow 0} \frac{1}{(\sqrt{1 + \sqrt{x+1}} + \sqrt{2})(\sqrt{x+1} + 1)}$$

We can now evaluate the limit continuously by direct substitution of $x = 0$:

$$\frac{1}{(\sqrt{1 + \sqrt{0+1}} + \sqrt{2})(\sqrt{0+1} + 1)}$$

$$= \frac{1}{(\sqrt{1+1} + \sqrt{2})(1+1)} = \frac{1}{(2\sqrt{2})(2)} = \frac{1}{4\sqrt{2}}$$

Rationalizing the denominator by multiplying top and bottom by $\sqrt{2}$ gives the final result:

$$\frac{\sqrt{2}}{8}$$

Monday 15 June 2026

5.1 Daily Limit

Limit of an Exponential Form of Hyperbolic and Trigonometric Functions

Evaluate the limit:

$$\lim_{x \rightarrow 0} \left(\frac{\cosh(x) + \cos(x)}{2} \right)^{\frac{1}{x^4}}$$

Solution: As $x \rightarrow 0$, both $\cosh(x)$ and $\cos(x)$ approach 1, making the base approach $\frac{1+1}{2} = 1$. The exponent $\frac{1}{x^4}$ approaches ∞ . This gives us the indeterminate form 1^∞ . Let L be the value of the limit. We proceed by taking the natural logarithm:

$$\ln(L) = \lim_{x \rightarrow 0} \frac{1}{x^4} \ln \left(\frac{\cosh(x) + \cos(x)}{2} \right)$$

We expand $\cosh(x)$ and $\cos(x)$ using their Maclaurin series near $x = 0$:

$$\cosh(x) = 1 + \frac{x^2}{2!} + \frac{x^4}{4!} + \frac{x^6}{6!} + \dots$$

$$\cos(x) = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots$$

When adding these two series together, the alternating signs of the x^2 , x^6 , and other x^{4k+2} terms neatly cancel each other out:

$$\cosh(x) + \cos(x) = 2 + 2\frac{x^4}{4!} + 2\frac{x^8}{8!} + \dots = 2 + \frac{2x^4}{24} + O(x^8)$$

Dividing the entire series by 2 yields:

$$\frac{\cosh(x) + \cos(x)}{2} = 1 + \frac{x^4}{24} + O(x^8)$$

Substitute this expansion back into the logarithmic limit expression:

$$\ln(L) = \lim_{x \rightarrow 0} \frac{1}{x^4} \ln \left(1 + \frac{x^4}{24} + O(x^8) \right)$$

Next, we use the standard Taylor series expansion $\ln(1 + u) = u - \frac{u^2}{2} + \dots$ where $u = \frac{x^4}{24} + O(x^8)$:

$$\ln \left(1 + \frac{x^4}{24} + O(x^8) \right) = \frac{x^4}{24} + O(x^8)$$

Substituting this back, the limit for $\ln(L)$ resolves algebraically:

$$\ln(L) = \lim_{x \rightarrow 0} \frac{1}{x^4} \left(\frac{x^4}{24} + O(x^8) \right) = \lim_{x \rightarrow 0} \left(\frac{1}{24} + O(x^4) \right) = \frac{1}{24}$$

Exponentiating both sides to solve for the original limit L gives the final answer:

$$L = e^{\frac{1}{24}}$$